SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO5: Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.(BTL 5)

Assessment Tool Weightage: 20%

Duration : 1 Hour QUESTIONS: 5 TotalMarks:50

Annexure A

Reverse Engineering

Code of Conduct:

I K. Reshma (192565062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

Introduction

A hospital uses an AI diagnosis assistant to analyze patient vital signs and laboratory reports. The assistant identifies high-risk patients who require immediate medical attention. By observing the system's behavior, we can reverse engineer the distributed AI pipeline and understand how patient data is collected, processed, scored, and used to generate alerts. PySpark RDDs help process large healthcare datasets in parallel, making diagnosis faster and more efficient.

1. Distributed AI Pipeline Using PySpark RDDs

The healthcare assistant works in a distributed pipeline. Patient information such as heart rate, blood pressure, oxygen level, temperature, and laboratory reports is collected from different hospital systems. These records are stored as RDDs in PySpark, where each patient record becomes one element. Data is divided across multiple nodes so thousands of records can be processed simultaneously. RDD transformations clean, filter, and prepare the data before analysis, while actions generate final outputs like risk scores and alerts.

- Pipeline Flow:
- Patient Records → Data Collection
- RDD Creation in PySpark
- Data Cleaning and Feature Extraction
- Risk Score Calculation
- Alert Generation
- Doctor Notification

2. Feature Extraction from Patient Records

Feature extraction converts raw medical records into meaningful values for prediction. Important features include age, blood pressure, heart rate, oxygen saturation, body temperature, blood sugar, hemoglobin level, white blood cell count, and medical history. Missing values are removed, units are standardized, and categorical values are converted into numerical form. These features become the input for the AI diagnosis model.

3. Risk Scoring Mechanism

The AI assistant assigns a risk score to each patient using extracted features. Patients with normal vital signs receive low scores, while patients with abnormal readings receive higher scores. For example, low oxygen levels, high fever, abnormal blood pressure, or very high blood sugar increase the score. The system classifies patients into Low Risk, Medium Risk, and High Risk categories to help doctors prioritize treatment.

4. Agent-Based Alert Mechanism

Different intelligent agents monitor patient conditions continuously. A monitoring agent collects updated vital signs, a scoring agent recalculates risk, an alert agent sends notifications when thresholds are crossed, and a reporting agent updates the patient dashboard. High-risk patients trigger immediate alerts to physicians and emergency staff.

5. Example Workflow

Step 1: Collect patient data. Step 2: Store records in PySpark RDDs. Step 3: Clean and preprocess the data. Step 4: Extract important health features. Step 5: Calculate risk score. Step 6: Generate alerts for high-risk patients. Step 7: Update dashboards and hospital records.

Advantages of Using PySpark RDDs

RDDs process large healthcare datasets quickly through parallel computing. They are fault tolerant, scalable, and suitable for real-time monitoring. Hospitals can analyze thousands of patient records efficiently and reduce delays in identifying critical patients.

Limitations

Prediction accuracy depends on data quality. Missing or incorrect records can affect results. AI alerts support doctors but should not replace medical judgment. Privacy and security of patient information are also important challenges.

Conclusion

Reverse engineering the healthcare diagnosis assistant shows how distributed AI systems process patient records using PySpark RDDs. Feature extraction converts raw medical data into useful inputs, risk scoring identifies patients needing urgent care, and agent-based alerts notify healthcare professionals immediately. This approach improves hospital efficiency, enables faster decision-making, and supports better patient care.